Manufacturing Process Tracking Platform

Software Engineer Preliminary Assignment | Pannipa Dangsomboon

1. Requirements Summary

The prototype enables users to create manufacturing orders, move each order through a defined sequence of production steps, view work in progress at every step, record resource consumption as work progresses, and obtain an overview of the current status of all orders. The implemented workflow is Material Prep, Machining, Quality Check, and Packaging. Orders retain their customer, product, quantity, priority, current state, and transition history.

2. Functional and Software Design

The user interface is a responsive React single-page application. It provides summary metrics, stage-based filtering, an order queue, resource-capacity indicators, a create-order form, and an advance-order dialog. The frontend communicates with a Python FastAPI JSON API. SQLAlchemy maps PostgreSQL entities for orders, steps, order history, resources, and consumption records. Docker Compose runs PostgreSQL, the API, and the web application as reproducible services.

New orders enter the first step automatically. An active order can advance only to the next sequential step; advancing after Packaging marks it complete. A transition can record multiple resource-consumption entries and closes the current history record in the same database transaction. API validation rejects invalid quantities, priorities, missing resources, and attempts to advance completed orders. Dashboard values are calculated from the underlying order and consumption records.

3. Simplified User Manual

Start Docker Desktop, run `docker compose up --build` from the repository root, and open `http://localhost:5173`. Select **New order**, enter the product, customer, quantity, and priority, then submit. The order appears in Material Prep. Select a stage card to filter the queue, or search by order code, product, or customer. To progress an active order, select the arrow at the end of its row, optionally enter resources consumed in the current stage, and confirm. Repeat until the order is completed. API documentation is available at `http://localhost:8000/docs`.

4. Software Summary

Meticuly Flow is a traceable manufacturing-control prototype built with React, FastAPI, SQLAlchemy, and PostgreSQL. Seed data makes the application immediately reviewable, while automated backend tests cover order creation, validation, stage advancement, history updates, and resource recording. The repository includes setup instructions, Docker configuration, source code, and this written summary. The prototype is intended for assessment and is not a production medical or manufacturing system.

5. AI-Assisted Development Disclosure

AI assistance was used to interpret the assignment, scaffold the application architecture, generate implementation code, prepare documentation, and review the core user journey. The resulting source code and behavior remain available for normal engineering review.